Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 38)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases .....

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Whether the numbers of dormice differ between woodlands with different management ..

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Whether mean dissolved oxygen differs between two streams .....

.....

Whether crop yield increases with fertiliser application (ranked data) .....

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0 1 . 2 Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 1 . 3 A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

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0 1 . 4 Explain the purpose of error bars on a graph of mean values.

[2 marks]

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0 2 . 1 Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

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0 2 . 2 Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

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0 2 . 3 Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

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Dissolved oxygen .....

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Water flow .....

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Day length .....

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Temperature .....

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0 3 . 1 Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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0 3 . 2 Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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0 3 . 3 Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

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2 .....

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3 .....

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0 4 Table 1 shows the number of plant species per m^2 in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m^2	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

0 4 . 1 Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

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0 4 . 2 Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

0 4 . 3 Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

0 4 . 4 Explain how burning maintains heathland.

[2 marks]

10

0 5 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 5 . 1 Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

1 .....

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2 .....

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0 5 . 2 Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 5 . 3 Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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0 6 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change

Ozone depletion

Tourism

Future mineral exploitation

0 6 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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0 7 Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 7 . 1 Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

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0 7 . 2 Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

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0 7 . 3 Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

1 .....

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2 .....

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0 7 . 4 Explain why bird wing bones are a useful model for engineers.

[1 mark]

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0 8 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 8 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 8 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

1 .....

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2 .....

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5

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0 9 . 1 Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability .....

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Dissolved oxygen .....

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pH .....

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0 9 . 2 Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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0 9 . 3 Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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1 0 . 1 Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

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Extra space

1 0 . 2 State what is meant by a **taxon**.

[1 mark]

10

Write an essay on one of the following topics.

1 1 . 1 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

1 1 . 2 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS